

Government Polytechnic, Osmanabad.

First Progressive Assessment Test of Theory AY-2023-24

Program Name: Computer Engineering Program (CO)

Semester : Third (CO3I)

Course Title : Data Structure using C - 22317

Marks : 20

Time: 1 Hour

Date : 04 / 09 /2023

Instructions:

- (1) All questions are compulsory.
- (2) Illustrate your answers with neat sketches wherever necessary.
- (3) Figures to the right indicate full marks.
- (4) Assume suitable data if necessary.
- (5) Preferably, write the answers in sequential order.

CO N	Course Outcome(CO)
a	Perform Basic Operations on Array
b	Apply different searching and sorting techniques.
c	Implement basic operations on stack and queue using Array representation.
d	Implement Basic Operations on Linked List.
e	Implement Program to create and traverse tree to solve problems.

Q.1 Attempt any FOUR.

08 Marks

QN	Question	Marks	CO
a	List Any Four Operations on Data Structure?	2	a
b	Define Overflow and Underflow w.r.t Stack?	2	c
c	Explain Time Complexity with example?	2	a
d	With Diagram, Show the Effect of following operations on stack with size 5? <i>Push(10),Push(20),Push(30),Pop(),Push(11),Pop(),Pop()</i>	2	c
e	Give classification on Data Structure?	2	a
f	Explain Concept of Recursion in data structure?	2	c

Q.2 Attempt any THREE.

12 Marks

QN	Question	Marks	CO
a	Explain algorithms for Push() and Pop()?	4	c
b	Write Algorithm for Binary Search?	4	b
c	Write a C Program for Insertion and Display operations on Array?	4	a
D	Write Algorithm for Bubble Sort?	4	b

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